

No m -cycles of the $3n-1$ map for $m \leq 58$

The Simons–de Weger template on the negative side, from a verification floor of 2^{51}

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Preprint. One external input, Rhin’s effective measure for linear forms in $\log 2$ and $\log 3$; one computational input, the laboratory’s own verification floor, a CPU certificate to 2^{44} and a GPU sweep to 2^{51} . Everything else is proved below or is exact arithmetic recorded with its margins, and every number in the tables is recomputed by a second, independent route (Section 8). Neither the $3n-1$ cycle question nor divergence on either map is settled here.

Abstract

Let $g(y) = y/2$ for even y and $g(y) = (3y-1)/2$ for odd y , the $3n-1$ map on the positive integers, equivalently the shortcut $3n+1$ map on the negative integers. Its known cycles are 1 , $(5, 7, 10)$ and the eleven-element cycle at 17 , and every start below 2^{51} reaches one of them. An m -cycle is a cycle with m maximal runs of odd elements. We transpose the template of Simons and de Weger for $3n+1$ m -cycles to this map, deriving its constants on this side rather than borrowing them: the odd step subtracts, so in the variable $u = y-1$ an odd run is exact multiplication by $3/2$, a run of a odd steps starts at $y \geq 2^a + 1$, the cycle equation bounds the linear form $\Lambda = o \log 3 - K \log 2$ by $m/(x_{\min} - 1)$ with constant one, and successive local minima obey $u_{i+1} < u_i^{\log_2 3}/2$. With Rhin’s bound this gives: the $3n-1$ map has no m -cycle with $1 \leq m \leq 58$ other than the two known ones. For $m \leq 52$ no admissible cycle length lies below Rhin’s ceiling; for $53 \leq m \leq 58$ the admissible lengths are excluded by the chaining, the closest by 9.8 bits. At $m = 59$ two lengths remain; they are the output of the template’s last step, not its input, and the floors that remove them are $2^{51.87}$ and $2^{55.18}$. The same tables give $m \leq 44$ from 2^{40} , $m \leq 49$ from 2^{44} , $m \leq 63$ from 2^{56} , $m \leq 68$ from 2^{60} and $m \leq 82$ from 2^{68} ; on the $3n+1$ side at the floor of Simons and de Weger the same enumeration and tests return their Lemma 18 to the unit. No published verification floor and no m -cycle theorem with $m \geq 3$ is known to us for this map.

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1. The map, the cycles, and what is known

Write g for the shortcut $3n-1$ map above. Its orbits on the positive integers are the orbits of the shortcut Collatz map $T(x) = x/2$, $(3x+1)/2$ on the negative integers under $x = -y$: $(3(-y)+1)/2 = -(3y-1)/2$. Three cycles are known,

$$(1), \quad (5, 7, 10), \quad (17, 25, 37, 55, 82, 41, 61, 91, 136, 68, 34),$$

with $(K, o) = (1, 1), (3, 2), (11, 7)$ steps and odd steps. Whether these are all the cycles, and whether every orbit is bounded, are the $3x-1$ questions of Lagarias’s bibliographies [L85], open.

The floor. Every $1 \leq y < 2^{51}$ reaches one of the three cycles. Below 2^{44} this is the laboratory’s CPU computation of 19–20 September 2026, descent by strong induction on odd starts in one hundred and twelve chunks: sixteen with a plain walker through 2^{40} (greatest step count 544

against a cap of 4000), then ninety-six chunks of $5 \cdot 2^{35}$ starts with a residue sieve modulo 2^{24} and a 2^{16} -step block map as an accelerator, 8246337208320 odd starts counted exactly (greatest step count 704 against a cap of 40000, counted in blocks of sixteen); the seven odd starts the first sixteen chunks had skipped, found from their own printed counts, are verified by full iteration. The range $[2^{44}, 2^{51})$ was verified on 21 September 2026 by a CUDA implementation of the same descent on one RTX 5090: the same sieve, then Barina’s domain switch [B21] with the sign flipped, which is Lemma 1 below, a whole odd run in one multiplication and a whole even run in one shift, on a 128-bit state with an overflow report. It was calibrated first on the CPU’s range, where it reproduces every count, both peaks and the plain walker’s step counts exactly, and cross-checked against the CPU walker on three 2^{37} windows inside the new range; 1117103813820416 odd starts, no failure, no new cycle, no overflow, greatest step count 847, peak about $2^{97.0}$, in 3517 seconds. No start failed and no new cycle appeared in either range. The verifier sources, the chunk reports, the calibration and the coverage checks are archived with the branch `negative_floor_3x1`. No published floor exists: the only statement in print is an unattributed comment on OEIS A037084, “up to at least 100000000”. The certificate is a computation and is labelled as one throughout.

m -cycles. A cycle with at least one even step splits into $m \geq 1$ maximal runs of odd elements alternating with runs of even elements. The first element of an odd run is a local minimum, the element after its last odd step a local maximum. The two known cycles with an even step have $m = 1$ and $m = 2$. Simons [S07] proves that g has exactly one nontrivial 2-cycle, at 17, and does so without any floor, by de Weger’s bounds on $0 < 3^K - 2^{K+L} < 3^{0.89K}$, Steiner’s 1-cycle theorem [St77] and an extremal inequality; he states that the floor-free route stops at $m \geq 3$. For $3n + 1$, Simons [Si05] excludes 2-cycles, Simons and de Weger [SdW] exclude m -cycles for $m \leq 68$ from the floor of November 2004, $X_0 = 301 \cdot 2^{50} > 3.3889 \cdot 10^{17}$, and Hercher [H23] reaches $m \leq 91$ from $695 \cdot 2^{60}$ with the arrangement of the valleys; Sinisalo [Si03] tabulates the cycle lengths a floor alone leaves on that side, the table this note’s m -free row transposes. The template of [SdW] has five steps: a floor; a bound on Λ from the cycle equation (their Lemma 4 and Corollary 5); a chaining of the local minima (Lemmas 6, 7); Crandall’s lemma generalized for a lower bound on the length (Lemma 10, Corollary 11) against Rhin’s bound for an upper one (Lemmas 12, 14), sharpened through the partial quotients of $\log_2 3$ (Lemma 16); and de Weger’s approximation lattice where a window remains (Lemma 18). At their floor the three stages give $m \leq 57$, $m \leq 63$ and $m \leq 68$. Nobody has run any of it on $3n - 1$, because the first step was missing; [S08], which extends the template to $3x + q$, $px + q$ and Guy’s permutation, skips the map. This note runs all five steps with the constants of this side. The exact enumeration of the admissible lengths by the three-gap walk is the list their approximation lattice produces ([SdW] Section 7: a two-dimensional lattice whose points of small norm are the pairs (K, L) in the window of Corollary 5 below the ceiling, found by a reduced basis and a search, each then “checked for fulfilling Corollary 5 and Lemma 7”); the two tests applied here to each length are those two. Run on the $3n + 1$ side at their floor, the machinery below returns their Lemma 18 (Section 5).

2. Notation

Throughout $x = \log 2 / \log 3$, $\delta = \log_2 3 = 1/x$. For a cycle C of g with K steps and o odd steps put

$$\Lambda(C) = o \log 3 - K \log 2.$$

For odd y put $u = y - 1$. The local minima of an m -cycle are $y_1, \dots, y_m$ in cyclic order, $u_i = y_i - 1$, the run from y_i has a_i odd steps, and $r_i \geq 1$ halvings follow it. $x_{\min}$ is the least element of C . The floor is $X_0 = 2^{51}$.

3. Four lemmas on the negative side

Lemma 1 (odd runs). Let $y \geq 3$ be odd, $u = y - 1$, $a = v_2(u)$. Then

$$g^k(y) - 1 = \left(\frac{3}{2}\right)^k (y - 1) \quad (0 \leq k \leq a),$$

$g^k(y)$ is odd for $k < a$ and $g^a(y)$ is even. Consequently a run of a odd steps starts at $y \geq 2^a + 1$, and its local maximum $M = g^a(y)$ satisfies $M \leq u^\delta + 1$.

Proof. For odd y , $g(y) - 1 = (3y - 1)/2 - 1 = \frac{3}{2}(y - 1)$. While the iterates stay odd this repeats, so $g^k(y) = 1 + 3^k u / 2^k$ as long as $g^j(y)$ is odd for $j < k$; and $1 + 3^k u / 2^k$ is an odd integer exactly when $2^{k+1} \mid u$, since 3^k is odd. Hence the first $a = v_2(u)$ iterates are odd and the next is even. Then $u \geq 2^a$ gives $y \geq 2^a + 1$, and $(3/2)^a \leq (3/2)^{\log_2 u} = u^{\log_2(3/2)} = u^{\delta-1}$ gives $M = 1 + (3/2)^a u \leq 1 + u^\delta$. $\square$

This is Hercher's Lemma 8 [H23] with $y \equiv 1 \pmod{2^a}$ in place of $y \equiv -1$: the floor $2^a + 1$ is larger by two, and it is attained at 5 ($a = 2$) and 17 ($a = 4$).

Lemma 2 (cycle equation). On a cycle C with at least one even step,

$$\Lambda(C) = - \sum_{y \in C \text{ odd}} \log\left(1 - \frac{1}{3y}\right) > 0, \quad \text{so} \quad 3^o > 2^K,$$

and $\Lambda(C) < \sum_{y \in C \text{ odd}} \frac{1}{3y - 1}$. The run from a local minimum y contributes less than $1/(y - 1)$. Hence, for an m -cycle,

$$\Lambda(C) < \sum_{i=1}^m \frac{1}{y_i - 1} \leq \frac{m}{x_{\min} - 1}.$$

Proof. An odd step multiplies by $\frac{3}{2}(1 - \frac{1}{3y})$ and an even step by $\frac{1}{2}$; around the cycle the product is 1, so $3^o 2^{-K} \prod_{\text{odd}} (1 - 1/(3y)) = 1$, which is the displayed identity, and the product is below 1. The bound $-\log(1 - t) \leq t/(1 - t)$ at $t = 1/(3y)$ gives $1/(3y - 1)$. Along the run from y , Lemma 1 gives $3g^k(y) - 1 = 3(3/2)^k u + 2 > 3(3/2)^k u$, so the run contributes less than $\frac{1}{3u} \sum_{k \geq 0} (2/3)^k = 1/u$. Finally the least element of C is odd (an even least element would halve to something smaller) and its predecessor is even (an odd predecessor y' would have $y' < (3y' - 1)/2 = x_{\min}$), so $x_{\min}$ is a local minimum and every $y_i \geq x_{\min}$. $\square$

This is [SdW] Lemma 4 and Corollary 5 with the sign flipped. The corresponding bound on the positive side, Hercher's Corollary 17, carries the constant 97/54, and [SdW] Lemma 4 has $\sum 1/x_i$ over the minima themselves. Here it is $1/(y_i - 1)$: the odd step subtracts, so the run's elements sit at or above the geometric progression rather than below it.

Lemma 3 (chaining). For consecutive local minima, $u_{i+1} < u_i^\delta/2$. Hence, with $B(m) = (\delta^m - 1)/(\delta - 1)$,

$$o = \sum_{i=1}^m a_i \leq B(m) \log_2 u_1 - \frac{B(m) - m}{\delta - 1}, \quad K < \delta o,$$

and therefore, taking $y_1 = x_{\min}$,

$$\log_2(x_{\min} - 1) > L_{\min}(K, m) := \frac{1}{B(m)} \left(\frac{K}{\delta} + \frac{B(m) - m}{\delta - 1} \right).$$

Proof. $y_{i+1} = M_i/2^{r_i}$ with $r_i \geq 1$ and $M_i \leq u_i^\delta + 1$ (Lemma 1), so $u_{i+1} = M_i/2^{r_i} - 1 \leq (u_i^\delta + 1)/2 - 1 < u_i^\delta/2$. Taking logarithms, $\log_2 u_{i+1} < \delta \log_2 u_i - 1$, and by induction $\log_2 u_i \leq \delta^{i-1} \log_2 u_1 - (\delta^{i-1} - 1)/(\delta - 1)$. Summing over i and using $a_i = v_2(u_i) \leq \log_2 u_i$ gives the bound on o ; $K \log 2 < o \log 3$ is $\Lambda > 0$. Solving for $\log_2 u_1$ gives $L_{\min}$. $\square$

The chaining is tight on the 17-cycle: $u_1 = 16$, $u_2 = 40$, and $16^\delta/2 = 40.5$. It is [SdW] Lemma 6 in the variable u : their $x_{i+1} < b^\delta x_i^\delta$ with $b = (1 + X_0^{-1})/2^{1/\delta}$ becomes exact here, with no X_0 in

the constant, because $u = y - 1$ is the conjugated variable itself. The bound on o is their Lemma 7 read as a lower bound on the least element rather than as an upper bound on Λ ; their constant $c_m = 2^{(m/\delta)(\delta-1)/(\delta^m-1)} b^{\delta/(\delta-1)-m/(\delta^m-1)}$ is, at $b^\delta = 1/2$, the $2^{-(B-m)/((\delta-1)B)}$ of Lemma 3.

Lemma 4 (admissible lengths). If an m -cycle has $x_{\min} \geq X_0$, then $0 < \Lambda < m/(X_0 - 1) < \log 3$, so $o = \lceil Kx \rceil$ and

$$\Lambda(K) = \log 3 (1 - \{Kx\}) < \frac{m}{X_0 - 1}.$$

The lengths K satisfying this are exactly those with $1 - \{Kx\} < m/((X_0 - 1) \log 3)$, and below any bound they are enumerated exactly by the three-gap walk: from the two least returns q_1, q_2 of $\{Kx\}$ into the two windows, found among the convergents and semiconvergents of x , every further member is the previous one plus q_1, q_2 or $q_1 + q_2$. The least member is Crandall's lower bound on the length.

Proof. Lemma 2 and $x_{\min} \geq X_0$. The walk is the classical three-distance structure of $\{Kx\}$ (Sós, Świerczkowski, Slater [3D]) and is the laboratory's `three_gap_walk`, checked against a direct scan at coarse thresholds. $\square$

[SdW] bound the same set from both ends before listing it: Lemma 10 and Corollary 11 give the least convergent denominator the floor admits, Lemma 16 caps the length through the largest partial quotient below that index, and Section 7 lists what lies between by an approximation lattice. The walk lists the set directly, and the two bounds are not needed.

4. Rhin's ceiling

Proposition 5. Rhin's Proposition [R87, p. 160, (7)] states that for integers u_0, u_1, u_2 with $H = \max(|u_1|, |u_2|) \geq 2$,

$$|u_0 + u_1 \log 2 + u_2 \log 3| \geq H^{-13.3},$$

with no further constant. For an m -cycle of length $K \geq 2$ with an even step, $(u_0, u_1, u_2) = (0, -K, o)$ and $H = K$, so

$$2^{L_{\min}(K, m)} < x_{\min} - 1 < \frac{m}{\Lambda} \leq m K^{13.3}.$$

Let $K_3(m)$ be the least integer such that $2^{L_{\min}(K, m)} \geq 2m K^{13.3}$ for every $K \geq K_3(m)$. Then every m -cycle has $K < K_3(m)$.

Proof. Lemmas 2 and 3 with Rhin's bound. The function $K \mapsto L_{\min}(K, m) - 13.3 \log_2 K - \log_2(2m)$ is convex, negative at $K = 2$, and has a larger root; beyond it the function increases, so $K_3(m)$ is well defined, and the doubled constant covers $m/\Lambda + 1$. $\square$

Rhin's bound is the one external input of this note. It is a theorem, not a hypothesis; it is external in the sense that nothing here re-proves it and no Lean statement carries it. [SdW] Lemma 12 states it as $\Lambda > e^{-13.3(0.46057 + \log K)}$ in their odd count K , which is (7) at $H = K + L < 1.000001 \delta K$ with $0.46057 = \log \delta$; the laboratory's Paper A and its cycle-gap branch carried that form, weaker by the factor $\delta^{13.3} = 457$ when H is the length itself, and it is (7) that is used here. The general two-logarithm bound of Laurent–Mignotte–Nesterenko [LMN] would replace 13.3 by an exponent in the thousands at these heights and is not competitive; Rhin's constant is specific to $\log 2$ and $\log 3$.

5. The theorem

Theorem 6. Let $X_0 = 2^{51}$. For every $1 \leq m \leq 58$ and every length $K < K_3(m)$ with $\Lambda(K) < m/(X_0 - 1)$,

$$2^{L_{\min}(K, m)} \geq \frac{m}{\Lambda(K)}.$$

Consequently, with Rhin's bound, the $3n - 1$ map has no m -cycle with $1 \leq m \leq 58$ whose least element is at least 2^{51} , and by the floor no m -cycle with $m \leq 58$ other than $(5, 7, 10)$ and the cycle at 17.

Proof. An m -cycle above the floor has $K < K_3(m)$ by Proposition 5, its length is admissible by Lemma 4, and Lemmas 2 and 3 give $2^{L_{\min}(K,m)} < x_{\min} - 1 \leq m/\Lambda(K)$, contradicting the display. The display is a finite exact computation: for each m the admissible lengths below $K_3(m)$ are enumerated by Lemma 4 and each is tested at eighty digits. For $m \leq 52$ there is no admissible length below $K_3(m)$ at all, so the conclusion there rests on Lemmas 2 and 4 and Proposition 5 alone. For $53 \leq m \leq 58$ the admissible lengths exist and fail the display by the margins of Table 1. $\square$

Table 1. Rows at the floor 2^{51} . “Margin” is $\max \log_2(m/(\Lambda(K) 2^{L_{\min}}))$ over the admissible $K < K_3(m)$, negative when every admissible length is excluded.

m	$m/(X_0 - 1)$	$K_3(m)$	admissible $K < K_3$	least admissible	margin (bits)
1	$4.4 \cdot 10^{-16}$	155	0	—	—
2	$8.9 \cdot 10^{-16}$	495	0	—	—
5	$2.2 \cdot 10^{-15}$	3926	0	—	—
10	$4.4 \cdot 10^{-15}$	57129	0	—	—
20	$8.9 \cdot 10^{-15}$	8393107	0	—	—
30	$1.3 \cdot 10^{-14}$	1094917645	0	—	—
40	$1.8 \cdot 10^{-14}$	134716197551	0	—	—
50	$2.2 \cdot 10^{-14}$	15974496522785	0	—	—
53	$2.4 \cdot 10^{-14}$	66574098182843	1	64789416887513	−549.2
54	$2.4 \cdot 10^{-14}$	107084917386112	2	64789416887513	−328.1
55	$2.4 \cdot 10^{-14}$	172208666666214	4	64789416887513	−188.5
56	$2.5 \cdot 10^{-14}$	276877898691766	6	64789416887513	−100.4
57	$2.5 \cdot 10^{-14}$	445072680198484	10	64789416887513	−44.8
58	$2.6 \cdot 10^{-14}$	715295665905327	16	64789416887513	−9.8
59	$2.6 \cdot 10^{-14}$	1149356600587428	27	64789416887513	+12.4

The closest exclusion, at $m = 58$, has 9.8 bits to spare at $K = 64789416887513$, the least admissible length at this floor; the same length is the first to survive at $m = 59$. At $m = 59$ the template leaves two lengths,

$$64789416887513, \quad 83130157078217,$$

with 12.4 and 5.0 bits of room; they are near-convergent lengths of x on the expanding side. They are the output of the template's last step, not its input: what removes each is a higher floor, at $2^{51.87}$ and $2^{55.18}$ respectively (Lemma 2's $x_{\min} - 1 < m/\Lambda$).

Table 2. What each floor buys, by the same tables.

floor	m excluded through	first open m	lengths left at the first open m
2^{40}	44	45	72448885240, 82888745831, 93328606422, 103768467013
2^{44}	49	50	539722056247, 757698850864, 975675645481, 1193652440098
2^{48}	49	50	1193652440098
2^{49}	54	55	9767196315401
2^{50}	56	57	28107936506105
2^{51}	58	59	64789416887513, 83130157078217

floor	m excluded through	first open m	lengths left at the first open m
2^{56}	63	64	766512153894657
$301 \cdot 2^{50}$ ([SdW]’s floor)	63	64	766512153894657
2^{60}	68	69	9881527843552324
2^{68}	82	83	6094436882695943503, 6724555128221608268

Two comparisons calibrate the method. First, on the $3n + 1$ side, with the window on the contracting side of x , the same enumeration and the same two tests at the floor of [SdW], $301 \cdot 2^{50}$, return their Lemma 18: no length for $64 \leq m \leq 68$, and at $69 \leq m \leq 72$ their five pairs (K, L) at the nine places of their table, with the floors that remove them to rounding: their first pair (5750934602875680, 3364081086781987) falls at 576.2, 584.6, 592.9, 601.3 times 2^{50} for $m = 69, \dots, 72$ against their 577, 585, 593, 602, their second at 623.6, 632.4 against 624, 633, and their three at $m = 72$ at 308.2, 666.8, 705.3 against 309, 667, 706. The same run lists one point more at $m = 72$, the double of their second pair, with 0.9 bits of room and a removing floor of $316.2 \cdot 2^{50}$; their table does not carry it, and the text does not say whether non-primitive pairs are discarded. On the $3n - 1$ side the same floor gives $m \leq 63$: the length 766512153894657 sits inside the window at $m = 64$ on the expanding side, where the contracting side has nothing below the ceiling until $m = 69$. The five values between 63 and 68 are the sign of the map, not a missing stage. Second, Hercher’s 91 at $695 \cdot 2^{60}$ against the plain template’s 82 at 2^{68} prices his valley arrangement at about nine values of m , the one refinement left to transpose. The floor enters only through the admissibility threshold, in steps: 2^{52} to 2^{55} buy nothing over 2^{51} ; 2^{56} buys $m \leq 63$, 2^{59} buys 64, 2^{60} buys 68.

6. The cycles that exist pass the same test

The inequalities were checked where they must not exclude. With the floor set at 17 and $m = 2$, the admissible lengths include 11 and it survives the display; with the floor at 5 and $m = 1$, length 3 survives; with the floor at 18, length 11 no longer appears, since the 17-cycle’s least element is below it. The cycle equation of Lemma 2 is verified in exact rationals on all three cycles, and Lemma 1’s run length and identity on every odd $y < 4000$.

7. What this note does not do

It excludes no Juggler cycle. The Juggler’s cycle words are, letter for letter, the words of the negative Collatz cycles ([A], Section 5.9), but a word shape does not transport a realization, so a $3n - 1$ m -cycle theorem constrains Juggler cycle words and nothing more. It does not settle the $3n - 1$ cycle question, m being bounded, and it says nothing about divergence on either map. Hercher’s refinement is not transposed; the lattice of admissible periods that Eliahou [E93] builds on the positive side is not transposed.

Lemmas 1 and 3 are machine-checked (`Problems.Collatz.NegativeMCycles`, twenty declarations, Lean 4 with Mathlib, no `sorry` and nothing off the kernel). The formalization writes the start of an odd run as $y = 2^a m + 1$ with m odd, which removes every truncated subtraction and puts the run in closed form, $g^k(2^a m + 1) = 3^k 2^{a-k} m + 1$ for $k \leq a$; Lemma 1’s four clauses and Lemma 3’s two halves are read off it, and the known cycles are checked against the same definitions inside the kernel. Formalizing it sharpened one hypothesis: the closed form needs no parity assumption on m , since the powers of two alone keep the state odd through the run, and oddness of m is used only to know that the run stops at a . Lemma 2 is not in Lean: its content is the cycle equation and the bound on Λ , which are real-analytic. Its first clause, $\Lambda > 0$, is the negative-cycle expansion already in Lean on the conjugate side (`neg_cycle_expanding`), together with the negative-side

finance and word shape (`neg_cycle_finance`, `neg_prefix_noncontracting`); the conjugation $x = -y$ that carries one to the other is stated in this note and is not itself formalized.

8. Verification

`python -m research.juggler_sequence.negative_m_cycles` recomputes both tables in a few seconds and writes `data/research/juggler/negative_m_cycles/summary.json`; `tests/research/juggler_sequence/test_negative_m_cycles.py` checks Lemma 1 on every odd $y < 4000$, the cycle equation on the three cycles, the tightness of Lemmas 1 and 3 at 17, that the known cycles pass, that K_3 is a ceiling, that the three-gap walk agrees with a direct scan, that the archived tables reproduce, and that the same machinery with the window on the contracting side returns [SdW] Lemma 18 at their floor. The floor’s certificate is with the branch `negative_floor_3x1`: the CPU chunk reports below 2^{44} , and above it the GPU sweep’s chunk reports, its calibration on the CPU’s range and the three spot-check windows (`python -m research.juggler_sequence.negative_floor_gpu calibrate, sweep 44 51, spot`). `lake build Problems.Collatz.NegativeMCycles` checks Lemmas 1 and 3; the module is in the Problems barrel, so the default build covers it, and `tests/research/juggler_sequence/test_negative_m_cycles_lean.py` holds it to no sorry, no `native_decide` and the three standard axioms.

```
Repository: https://github.com/sneakyweasel/btlab
Commit:    d6e8a4ae20a5c82f46fc5e673a1bf8d1a344e29c
Lean:      leanprover/lean4:v4.33.1
Mathlib:   v4.33.1 (lake-manifest rev 0df444a360eaa60ab8c11dca51a86af692955474)
Build:     lake build Problems.Collatz.NegativeMCycles (from formal/)
Tables:    python -m research.juggler_sequence.negative_m_cycles
Floor:     python -m research.juggler_sequence.negative_floor_gpu sweep 44 51
Recheck:   python tools/check_3n_minus_1_note_numeric.py
```

The commit is the repository state that produced the tables, the floor records and the Lean module; every file those commands read or write is byte-identical there to the version this paper reports. A later editorial commit of this text, including the one that adds the build tooling for this paper, does not change them.

Acknowledgments and use of AI

Large language models assisted the development of this work, including prose, proposed proof arguments, the Lean formalization, the CUDA verifier, and the computations. The sources of Simons–de Weger 2005 and Rhin 1987 were read directly and the constants taken from them, not from memory; the transposition to the $3n - 1$ side, the enumeration, the tables and their independent recomputation were produced and checked in that collaboration. AI assistance is not independent mathematical validation. The author is responsible for the statements, the proofs, the code, and the decision to make this version public.

Availability and version

This is version 1.0.0 of Paper D, of 21 September 2026, at the floor 2^{51} . It is a preprint and has not been refereed. The manuscript, the probe that computes the tables, the independent check that recomputes them, the Lean module, the CUDA verifier and the floor certificate with its chunk reports are in the repository `sneakyweasel/btlab` at the commit named in Section 8; `python tools/build_paper_d.py` rebuilds this document from its canonical Markdown and `--check` verifies every generated copy against the manifest. The verification floor is a computation of this laboratory, not a result from the literature, and is labelled as one throughout; a higher floor raises the theorem’s m and would be a new version rather than a correction of this one.

References

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